🔹 **ASP.NET Core MVC – 50 Interview Questions & Answers**

✅ 1. Basics of ASP.NET Core MVC

1. What is ASP.NET Core MVC?
ASP.NET Core MVC is a lightweight, open-source web framework from Microsoft used to build web applications using the Model-View-Controller pattern.

2. What is MVC pattern?
MVC stands for:

- Model → Data & business logic
- View → UI
- Controller → Handles request & response

3. Why use MVC?

- Separation of concerns
- Testability
- Maintainability
- Scalability

4. What is ASP.NET Core?
A cross-platform, high-performance framework to build web apps, APIs, and microservices.

5. Difference between ASP.NET MVC and ASP.NET Core MVC?

- Core is cross-platform
- Better performance
- Built-in dependency injection
- No System.Web dependency

✅ 2. Controllers

6. What is a Controller?
A class that handles HTTP requests and returns responses.

7. How to create a controller?
Inherit from Controller class or use [Controller] attribute.

8. What is Action Method?
A public method in controller that handles a request.

9. What are Action Results?
Return types like:

- ViewResult
- JsonResult
- RedirectResult

10. Difference between IActionResult and ActionResult?

- IActionResult → Interface
- ActionResult → Concrete class

✅ 3. Models

11. What is a Model?
Represents application data and business logic.

12. What is ViewModel?
A model specifically designed to pass data to views.

13. Difference between Model and ViewModel?

- Model → Database entity
- ViewModel → UI-specific data

14. What are Data Annotations?
Attributes used for validation like [Required], [StringLength].

15. Example of validation attribute?

```
[Required]

public string Name { get; set; }
```

✅ 4. Views

16. What is a View?
UI component that displays data.

17. What is Razor View Engine?
Syntax used to embed C# in HTML.

18. What is @model in Razor?
Defines the model type for the view.

19. Difference between ViewBag and ViewData?

- ViewBag → Dynamic
- ViewData → Dictionary

20. What is Partial View?
Reusable UI component.

✅ 5. Routing

21. What is Routing?
Maps URL to controller and action.

22. Types of routing?

- Conventional
- Attribute routing

23. Example of attribute routing?

[Route("home/index")]

24. What is default route?

{controller=Home}/{action=Index}/{id?}

25. What is endpoint routing?
Modern routing system introduced in ASP.NET Core.

✅ 6. State Management

26. What is State Management?
Maintaining user data across requests.

27. Types of state management?

- Session
- Cookies
- TempData

28. What is Session?
Stores user data on server.

29. What is TempData?
Stores data temporarily between requests.

30. Difference between Session and TempData?

- Session → persists longer
- TempData → one-time use

✅ 7. Dependency Injection

31. What is Dependency Injection (DI)?
Design pattern to inject dependencies into classes.

32. Where is DI configured?
In Program.cs

33. Types of lifetimes?

- Transient
- Scoped
- Singleton

34. What is Scoped lifetime?
Created once per request.

✅ 8. Entity Framework Core

35. What is EF Core?
ORM framework for database operations.

36. What is DbContext?
Main class to interact with database.

37. What is DbSet?
Represents table in database.

38. What is Migration?
Process of updating database schema.

39. Code First vs Database First?

- Code First → model → DB
- DB First → DB → model

✅ 9. Forms & Tag Helpers

40. What are Tag Helpers?
Server-side helpers to generate HTML.

41. Example of Tag Helper?

```
<input asp-for="Name" />
```

42. What is HTML Helper?
Used to generate HTML via C# code.

43. Difference between Tag Helper and HTML Helper?

- Tag Helper → HTML-like
- HTML Helper → C# syntax

✅ 10. Filters & Middleware

44. What are Filters?
Used to run code before/after action execution.

45. Types of Filters?

- Authorization
- Action
- Result

- Exception

46. What is Middleware?
Handles request pipeline.

✅ 11. Security

47. What is Authentication?
Verifying user identity.

48. What is Authorization?
Checking user permissions.

49. What is Anti-Forgery Token?
Prevents CSRF attacks.

✅ 12. Miscellaneous

50. What is Kestrel?
Cross-platform web server used in ASP.NET Core.